

Articles - Regression Analysis

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Regression analysis (or **regression model**) consists of a set of *machine learning* methods that allow us to predict a continuous outcome variable (y) based on the value of one or multiple predictor variables (x).

Briefly, the goal of regression model is to build a mathematical equation that defines y as a function of the x variables. Next, this equation can be used to predict the outcome (y) on the basis of new values of the predictor variables (x).

Linear regression is the most simple and popular technique for predicting a continuous variable. It assumes a linear relationship between the outcome and the predictor variables. See Chapter [@ref\(linear-regression\)](#).

The linear regression equation can be written as $y = b_0 + b \cdot x$, where:

- b_0 is the intercept,
- b is the regression weight or coefficient associated with the predictor variable x .

Technically, the linear regression coefficients are determined so that the error in predicting the outcome value is minimized. This method of computing the beta coefficients is called the **Ordinary Least Squares** method.

When you have multiple predictor variables, say x_1 and x_2 , the regression equation can be written as $y = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2$. In some situations, there might be an **interaction effect** between some predictors, that is for example, increasing the value of a predictor variable x_1 may increase the effectiveness of the predictor x_2 in explaining the variation in the outcome variable. See Chapter [@ref\(interaction-effects-in-multiple-regression\)](#).

Note also that, linear regression models can incorporate both continuous and **categorical predictor variables**. See Chapter [@ref\(regression-with-categorical-variables\)](#).

When you build the linear regression model, you need to **diagnostic** whether linear model is suitable for your data. See Chapter [@ref\(regression-assumptions-and-diagnostics\)](#).

In some cases, the relationship between the outcome and the predictor variables is not linear. In these situations, you need to build a **non-linear regression**, such as *polynomial* and *spline regression*. See Chapter [@ref\(polynomial-and-spline-regression\)](#).

When you have multiple predictors in the regression model, you might want to select the best combination of

predictor variables to build an optimal predictive model. This process called **model selection**, consists of comparing multiple models containing different sets of predictors in order to select the best performing model that minimize the prediction error. Linear model selection approaches include **best subsets regression** (Chapter @ref(best-subsets-regression)) and **stepwise regression** (Chapter @ref(stepwise-regression))

In some situations, such as in genomic fields, you might have a large multivariate data set containing some correlated predictors. In this case, the information, in the original data set, can be summarized into few new variables (called principal components) that are a linear combination of the original variables. This few principal components can be used to build a linear model, which might be more performant for your data. This approach is known as **principal component-based methods** (Chapter @ref(pcr-and-pls-regression)), which include: **principal component regression** and **partial least squares regression**.

An alternative method to simplify a large multivariate model is to use **penalized regression** (Chapter @ref(penalized-regression)), which penalizes the model for having too many variables. The most well known penalized regression include **ridge regression** and the **lasso regression**.

You can apply all these different regression models on your data, compare the models and finally select the best approach that explains well your data. To do so, you need some statistical metrics to compare the performance of the different models in explaining your data and in predicting the outcome of new test data.

The best model is defined as the model that has the lowest prediction error. The most popular metrics for comparing regression models, include:

- **Root Mean Squared Error**, which measures the model prediction error. It corresponds to the average difference between the observed known values of the outcome and the predicted value by the model. RMSE is computed as $RMSE = \text{mean}((\text{observeds} - \text{predicted})^2)^{0.5}$. The lower the RMSE, the better the model.
- **Adjusted R-square**, representing the proportion of variation (i.e., information), in your data, explained by the model. This corresponds to the overall quality of the model. The higher the adjusted R², the better the model

Note that, the above mentioned metrics should be computed on a new test data that has not been used to train (i.e. build) the model. If you have a large data set, with many records, you can randomly split the data into training set (80% for building the predictive model) and test set or validation set (20% for evaluating the model performance).

One of the most robust and popular approach for estimating a model performance is **k-fold cross-validation**. It can be applied even on a small data set. k-fold cross-validation works as follow:

1. Randomly split the data set into k-subsets (or k-fold) (for example 5 subsets)
2. Reserve one subset and train the model on all other subsets
3. Test the model on the reserved subset and record the prediction error
4. Repeat this process until each of the k subsets has served as the test set.
5. Compute the average of the k recorded errors. This is called the cross-validation error serving as the performance metric for the model.

Taken together, the best model is the model that has the lowest cross-validation error, RMSE.

In this Part, you will learn different methods for regression analysis and we'll provide practical example in **R**. The following techniques are described:

- Ordinary least squares (Chapter [@ref\(linear-regression\)](#))
 - Simple linear regression
 - Multiple linear regression
- Model selection methods:
 - Best subsets regression (Chapter [@ref\(best-subsets-regression\)](#))
 - Stepwise regression (Chapter [@ref\(stepwise-regression\)](#))
- Principal component-based methods (Chapter [@ref\(pcr-and-pls-regression\)](#)):
 - Principal component regression (PCR)
 - Partial least squares regression (PLS)
- Penalized regression (Chapter [@ref\(penalized-regression\)](#)):
 - Ridge regression
 - Lasso regression

Examples of data set

We'll use three different data sets: `marketing` [datarium package], the built-in R `swiss` data set, and the `Boston` data set available in the `MASS` R package.

marketing data

The `marketing` data set [datarium package] contains the impact of three advertising medias (youtube, facebook and newspaper) on sales. It will be used for predicting sales units on the basis of the amount of money spent in the three advertising medias.

Data are the advertising budget in thousands of dollars along with the sales. The advertising experiment has been repeated 200 times with different budgets and the observed sales have been recorded.

First install the `datarium` package:

```
if(!require(devtools)) install.packages("devtools")
devtools::install_github("kassambara/datarium")
```

Then, load `marketing` data set as follow:

```
data("marketing", package = "datarium")
head(marketing, 3)
```

swiss data

The `swiss` describes 5 socio-economic indicators observed around 1888 used to predict the fertility score of 47 swiss French-speaking provinces.

Load and inspect the data:

```
data("swiss")
head(swiss, 3)
```

```
##      Fertility Agriculture Examination Education Catholic
## Courtelary      80.2      17.0      15      12      9.96
## Delemont       83.1      45.1       6       9      84.84
## Franches-Mnt   92.5      39.7       5       5      93.40
##      Infant.Mortality
## Courtelary      22.2
## Delemont        22.2
## Franches-Mnt    20.2
```

The data contain the following variables:

- Fertility Ig: common standardized fertility measure
- Agriculture: % of males involved in agriculture as occupation
- Examination: % draftees receiving highest mark on army examination
- Education: % education beyond primary school for draftees.
- Catholic: % 'catholic' (as opposed to 'protestant').
- Infant.Mortality: live births who live less than 1 year.

Boston data

Boston [in **MASS** package] will be used for predicting the median house value (**medv**), in Boston Suburbs, using different predictor variables:

- **crim** , per capita crime rate by town
- **zn** , proportion of residential land zoned for lots over 25,000 sq.ft
- **indus** , proportion of non-retail business acres per town
- **chas** , Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
- **nox** , nitric oxides concentration (parts per 10 million)
- **rm** , average number of rooms per dwelling
- **age** , proportion of owner-occupied units built prior to 1940
- **dis** , weighted distances to five Boston employment centres
- **rad** , index of accessibility to radial highways
- **tax** , full-value property-tax rate per USD 10,000
- **ptratio** , pupil-teacher ratio by town
- **black** , $1000(B - 0.63)^2$ where B is the proportion of blacks by town
- **lstat** , percentage of lower status of the population
- **medv** , median value of owner-occupied homes in USD 1000's

Load and inspect the data:

```
data("Boston", package = "MASS")
head(Boston, 3)

##      crim zn indus chas  nox  rm  age  dis rad tax ptratio black lstat
## 1 0.00632 18  2.31   0 0.538 6.58 65.2 4.09  1 296   15.3  397  4.98
## 2 0.02731  0  7.07   0 0.469 6.42 78.9 4.97  2 242   17.8  397  9.14
## 3 0.02729  0  7.07   0 0.469 7.18 61.1 4.97  2 242   17.8  393  4.03
##      medv
## 1 24.0
## 2 21.6
## 3 34.7
```